CORE PURE / PROOF BY INDUCTION

Proof by Induction - Series

Mark scheme 10 questions · 73 marks

Questions

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	Total	**73**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- [$\boldsymbol{n}$] ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of $\boldsymbol{g}$

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$n=1$: LHS $=1\times 2=2$, RHS $=2+(1-1)2^2=2$	**B1**	Both sides evaluated at $n=1$ and shown equal to 2.
	Assume $\sum_{r=1}^{k} r\,2^r = 2+(k-1)2^{k+1}$ $\sum_{r=1}^{k+1} r\,2^r = \sum_{r=1}^{k} r\,2^r + (k+1)2^{k+1}$	**M1**	States the assumption for $n=k$ and adds a $(k+1)$th term of the form $(k+1)2^{\cdots}$ to the sum to k. Condone a slip in the power.
	$=2+(k-1)2^{k+1}+(k+1)2^{k+1}$ $=2+\big((k-1)+(k+1)\big)2^{k+1}$	**M1**	Replaces the sum to k by the hypothesis and collects the terms in 2^{k+1}, or writes both with a common power of 2. Condone sign slips.
	$=2+2k\,2^{k+1}$	**A1**	oe, such as $2+k\,2^{k+2}$.
	$=2+k\,2^{k+2}=2+\big((k+1)-1\big)2^{(k+1)+1}$	**A1**	cso. $2+\big((k+1)-1\big)2^{(k+1)+1}$ seen, or $2+k\,2^{k+2}$ explicitly compared with the result at $n=k+1$.
	True for $n=1$; true for $n=k \Rightarrow$ true for $n=k+1$; so true for all $n\geq 1$	**E1**	States all three points: true for $n=1$, true for $n=k$ implies true for $n=k+1$, so true for all $n\geq 1$. One sentence accepted.
	Total for Question 1	**6**	

Question	Scheme	Marks	Notes
2	$n = 1$: LHS $= 2 + 2 \times 3 = 8$ RHS $= 2^2 + 3^2 - 5 = 8$	**B1**	Both sides evaluated to 8; a bare $8 = 8$ is not enough.
	Assumes true for $n = k$: $\sum_{r=1}^{k+1} = 2^{k+1} + 3^{k+1} - 5$ $+ 2^{k+1} + 2 \times 3^{k+1}$	**M1**	States the assumption for $n = k$ and adds a $(k+1)$th term to its RHS, or shows RHS$(k+1)$ − RHS(k) is that term. Condone a slip in the term.
	$= 2 \times 2^{k+1} + 3 \times 3^{k+1} - 5$	**M1 A1**	**M1** Collects like powers of 2 and of 3 from their sum, e.g. $2^{k+1} + 2^{k+1}$ to $2 \times 2^{k+1}$. **A1** Correct, oe.
	$= 2^{k+2} + 3^{k+2} - 5$	**A1**	cso; reached from their sum, or $2^{k+2} + 3^{k+2} - 5$ shown equal to their expression; A0 if asserted.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k + 1$; so true for all $n \geq 1$	**E1**	States all three: true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, hence true for all $n \geq 1$.
	Total for Question 2	**6**	

Question	Scheme	Marks	Notes
3	$n = 1$: LHS $= 1(2)(3) = 6$ RHS $= \frac{1}{4}(1)(2)(3)(4) = 6$	**B1**	Both sides evaluated at $n = 1$; "true for $n = 1$" alone is not enough.
	Assumes true for $n = k$: $\sum_{r=1}^{k+1} r(r+1)(r+2)$ $= \frac{1}{4}k(k+1)(k+2)(k+3) + (k+1)(k+2)(k+3)$	**M1**	States the assumption for $n = k$, in words or symbols, and adds the $(k+1)$th term $(k+1)(k+2)(k+3)$ to the sum to k.
	$= (k+1)(k+2)(k+3)\left(\frac{k}{4} + 1\right)$	**M1 A1**	**M1** Uses the hypothesis and takes out a common factor of at least two of $k+1$, $k+2$, $k+3$, or expands both terms into one quartic. Condone slips in the remaining bracket. **A1** Correct factorised expression oe, or $\frac{1}{4}(k^4 + 10k^3 + 35k^2 + 50k + 24)$.
	$= \frac{1}{4}(k+1)(k+2)(k+3)(k+4)$	**A1**	cso, oe in terms of $k+1$. A quartic must be shown equal to the target with $n = k+1$.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k+1$; so true for all positive integers n	**E1**	States all three points: true for $n = 1$, true for $n = k$ implies true for $n = k+1$, hence true for all positive integers n.
	Total for Question 3	**6**	

Question	Scheme	Marks	Notes
4	$n = 1$: LHS $= 1 \times 3^0 = 1$ RHS $= \frac{1+(2-1)3^1}{4} = 1$	**B1**	Both sides shown to equal 1; a bare statement that it is true for $n = 1$ is not enough.
	Assume true for $n = k$: $\sum_{r=1}^{k+1} r\,3^{r-1}$ $= \sum_{r=1}^{k} r\,3^{r-1} + (k+1)3^k$	**M1**	States the assumption for $n = k$ and adds a $(k+1)$th term of the form $(k+1)3^p$ to the sum to k. Condone a slip in the power.
	$= \frac{1+(2k-1)3^k+4(k+1)3^k}{4}$	**M1**	Replaces the sum to k by the hypothesis and writes the result as a single fraction over 4 or a multiple of 4. Condone sign slips.
	$= \frac{1+(6k+3)3^k}{4}$	**A1**	Correct numerator collected as one multiple of 3^k, oe.
	$= \frac{1+(2k+1)3^{k+1}}{4}$	**A1**	cso. Reached by factorising 3^{k+1}, or by simplifying the RHS at $n = k+1$ to the same expression as their sum.
	True for $n = 1$; true for $n = k$ $\Rightarrow$ true for $n = k+1$ $\Rightarrow$ true for all $n \in \mathbb{Z}^+$	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k+1$, and hence true for all positive integers n; any order.
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5(a)	$n = 1$: LHS $= 1^3 = 1$, RHS $= \frac{1}{4}(1)^2(2)^2 = 1$	**B1**	Both sides shown equal to 1.
	Assume $\sum_{r=1}^{k} r^3 = \frac{1}{4}k^2(k+1)^2$ $\sum_{r=1}^{k+1} r^3 = \sum_{r=1}^{k} r^3 + (k+1)^3$	**M1**	Assumes the result for $n = k$ and adds $(k+1)^3$ to the sum to k.
	$= \frac{1}{4}k^2(k+1)^2 + (k+1)^3$ $= (k+1)^2(\ldots)$	**M1**	Substitutes the hypothesis and takes out a factor $(k+1)^2$ or $\frac{1}{4}(k+1)^2$; or expands to a quartic for comparison with $\frac{1}{4}(k+1)^2(k+2)^2$ expanded.
	$= (k+1)^2\left(\frac{k^2}{4} + k + 1\right)$	**A1**	Correct expression, oe such as $\frac{1}{4}(k^4 + 6k^3 + 13k^2 + 12k + 4)$.
	$= \frac{1}{4}(k+1)^2(k+2)^2$	**A1**	Reaches $\frac{1}{4}(k+1)^2(k+2)^2$, or shows it equals their correct quartic, without error.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ so true for all integers $n \geq 1$	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k + 1$, hence true for all $n \geq 1$; may be split across the proof.
		[6]	
5(b)	$\frac{1}{4}(2n)^2(2n+1)^2 = n^2(2n+1)^2$	**B1**	Or any equivalent factorised form. Not an expanded polynomial.
		[1]	
5(c)	$\sum_{r=1}^{n}(2r)^3 = 8\sum_{r=1}^{n} r^3 = 8 \times \frac{1}{4}n^2(n+1)^2$	**M1**	Takes out the factor 8 or 2^3 and applies the formula of part (a) with upper limit n.
	$= 2n^2(n+1)^2$	**A1**	Or any equivalent factorised form.
		[2]	
5(d)	First $2n$ cubes − even cubes: $1^3 + 3^3 + \cdots + (2n-1)^3$ $= n^2(2n+1)^2 - 2n^2(n+1)^2$	**M1 A1**	**M1** Subtracts their (c) from their (b); or expands $\sum(2r-1)^3$ and applies standard summation formulae. **A1** Correct difference, or correct unsimplified sum by the other route.
	$= n^2(2n^2 - 1)$, $a = 1$, $b = 2$	**A1**	cso. $a = 1$, $b = 2$ stated, or $n^2(2n^2 - 1)$ reached.
		[3]	
	Total for Question 5	**12**	

Question	Scheme	Marks	Notes
6(a)	$n=1$: LHS $=\frac{1}{1\times3}=\frac{1}{3}$, RHS $=\frac{1}{2+1}=\frac{1}{3}$	**B1**	Both sides evaluated as $\frac{1}{3}$; a bare statement that it is true for $n=1$ is not enough.
	Assume true for $n=k$: $\sum_{r=1}^{k+1}=\frac{k}{2k+1}+\frac{1}{(2k+1)(2k+3)}$	**M1**	States the assumption for $n=k$ and adds the $(k+1)$th term to the sum of k terms. The term need not be simplified.
	$=\frac{k(2k+3)+1}{(2k+1)(2k+3)}$	**M1 A1**	**M1** Replaces the sum of k terms by $\frac{k}{2k+1}$ and combines over any common denominator. Condone slips in the numerator. **A1** Any correct single fraction, unsimplified.
	$=\frac{(2k+1)(k+1)}{(2k+1)(2k+3)}=\frac{k+1}{2(k+1)+1}$	**A1**	cso. Correct factorisation reaching $\frac{k+1}{2k+3}$ or $\frac{k+1}{2(k+1)+1}$.
	True for $n=1$; true for $n=k\Rightarrow$ true for $n=k+1$; so true for all positive integers n	**E1**	States all three points: true for $n=1$, true for $n=k$ implies true for $n=k+1$, hence true for all positive integers n.
		[6]	
6(b)	$\sum_{r=n+1}^{2n}=\frac{2n}{4n+1}-\frac{n}{2n+1}$ $=\frac{2n(2n+1)-n(4n+1)}{(4n+1)(2n+1)}$	**M1**	Replaces n by $2n$ in the result of (a), subtracts the sum of n terms and combines over a common denominator. Condone slips in the numerator.
	$=\frac{n}{(4n+1)(2n+1)}$ $a=4,\ b=1,\ c=2,\ d=1$	**A1**	cso. Factors in either order, so $a=2,\ b=1,\ c=4,\ d=1$ also scores.
		[2]	
	Total for Question 6	**8**	

Question	Scheme	Marks	Notes
7	$n = 1$: LHS $= \frac{1}{2}$, RHS $= 2 - \frac{3}{2} = \frac{1}{2}$	**B1**	Both sides shown equal to $\frac{1}{2}$; not for one side alone.
	Assume $\sum_{r=1}^{k} \frac{r}{2^r} = 2 - \frac{k+2}{2^k}$ $\sum_{r=1}^{k+1} \frac{r}{2^r} = 2 - \frac{k+2}{2^k} + \frac{k+1}{2^{k+1}}$	**M1**	States or uses the result for $n = k$ and adds a $(k+1)$th term to it. Condone a slip in the term.
	$= 2 - \frac{2(k+2)-(k+1)}{2^{k+1}}$	**M1 A1**	**M1** Combines their expression over a common denominator such as 2^{k+1}. Condone sign slips. **A1** Correct single fraction oe; need not be simplified.
	$= 2 - \frac{(k+1)+2}{2^{k+1}}$	**A1**	cso. $2 - \frac{k+3}{2^{k+1}}$ scores only if linked to the result for $n = k+1$, stated as target or conclusion.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k + 1$ so true for all $n \geq 1$	**E1**	States true for $n = 1$, truth for $n = k$ implies truth for $n = k + 1$, hence true for all $n \geq 1$. Needs the assumption for $n = k$ stated somewhere.
	Total for Question 7	**6**	

Question	Scheme	Marks	Notes
8	$n=1$: LHS $=\frac{1}{1\times2\times3}=\frac{1}{6}$ RHS $=\frac{1\times4}{4\times2\times3}=\frac{1}{6}$	**B1**	Both sides evaluated as $\frac{1}{6}$. Not for a bare "true for $n=1$".
	Assume $\sum_{r=1}^{k}\frac{1}{r(r+1)(r+2)}=\frac{k(k+3)}{4(k+1)(k+2)}$ $\sum_{r=1}^{k+1}\frac{1}{r(r+1)(r+2)}$ $=\sum_{r=1}^{k}\frac{1}{r(r+1)(r+2)}+\frac{1}{(k+1)(k+2)(k+3)}$	**M1**	States the assumption for $n=k$ and adds the $(k+1)$th term to the sum to k terms. Condone a slip in the added term.
	$=\frac{k(k+3)}{4(k+1)(k+2)}+\frac{1}{(k+1)(k+2)(k+3)}$ $=\frac{\ldots}{4(k+1)(k+2)(k+3)}$	**M1**	Replaces the sum to k terms by the hypothesis and combines with their $(k+1)$th term over a common denominator. Condone numerator slips.
	$=\frac{k(k+3)^2+4}{4(k+1)(k+2)(k+3)}$	**A1**	Any correct single fraction, unsimplified or with the numerator expanded.
	$=\frac{(k+1)^2(k+4)}{4(k+1)(k+2)(k+3)}$ $=\frac{(k+1)(k+4)}{4(k+2)(k+3)}$	**A1**	cso, oe in terms of $k+1$: numerator factorised, or the $n=k+1$ result shown equal to their fraction; A0 if asserted.
	True for $n=1$; true for $n=k\Rightarrow$ true for $n=k+1$ $\Rightarrow$ true for all $n\in\mathbb{Z}^+$	**E1**	Concludes with all three points: true for $n=1$; if true for $n=k$ then true for $n=k+1$; so true for all $n\in\mathbb{Z}^+$.
	Total for Question 8	**6**	

Question	Scheme	Marks	Notes
9(a)	$n = 1$: LHS $= 1^3 = 1$, RHS $= 1^2(2-1) = 1$	**B1**	Both sides shown equal to 1; a bare "true for $n = 1$" is not enough.
	Assumes $\sum_{r=1}^{k}(2r-1)^3 = k^2(2k^2-1)$ $\sum_{r=1}^{k+1}(2r-1)^3 = \sum_{r=1}^{k}(2r-1)^3 + (2k+1)^3$	**M1**	Assumes the result for $n = k$ and adds the $(k+1)$th term, $(2k+1)^3$ or $(2(k+1)-1)^3$, to the sum to k.
	$= k^2(2k^2-1) + (2k+1)^3 = \ldots$	**M1**	Substitutes $k^2(2k^2-1)$ for the sum to k and expands or factorises towards the $n = k+1$ form. Condone slips in expanding the cube.
	$= 2k^4 + 8k^3 + 11k^2 + 6k + 1$	**A1**	oe, such as a correct factorised form.
	$(k+1)^2(2(k+1)^2-1) = (k^2+2k+1)(2k^2+4k+1)$ $= 2k^4 + 8k^3 + 11k^2 + 6k + 1$	**A1**	Shows the quartic equals $(k+1)^2(2(k+1)^2-1)$, by expanding this or factorising the quartic, without error.
	True for $n = 1$; true for $n = k \Rightarrow$ true for $n = k+1$ $\Rightarrow$ true for all positive integers n	**E1**	States true for $n = 1$, true for $n = k$ implies true for $n = k+1$, and so true for all positive integers n; may be spread through the working.
		[6]	
9(b)	$\sum_{r=1}^{2n}(2r-1)^3 = (2n)^2\left(2(2n)^2-1\right)$	**M1**	Replaces n by $2n$ throughout the formula from (a). Need not be simplified.
	$= 4n^2(8n^2-1)$	**A1**	oe, such as $32n^4 - 4n^2$.
	"$4n^2(8n^2-1)$" $= kn^2(2n^2-1)$	**M1**	Sets their sum to $2n$, obtained from (a)'s formula, equal to k times $n^2(2n^2-1)$.
	$32n^2 - 4 = 2kn^2 - k \Rightarrow (32-2k)n^2 = 4-k$	**dM1**	Divides by n^2 and collects the n^2 terms to reach the form $n^2(\ldots) = \ldots$. Condone sign slips.
	$n^2 = \frac{k-4}{2(k-16)}$	**A1 AG**	Correct substituted equation leading without error to the printed form; combined steps accepted.
		[5]	
	Total for Question 9	**11**	

Question	Scheme	Marks	Notes
10	$n=1$: LHS $= \log\left(1-\frac{1}{4}\right) = \log\frac{3}{4}$ RHS $= \log\left(\frac{2!\,3!}{2^4}\right) = \log\frac{3}{4}$	**B1**	Both sides shown equal to $\log\frac{3}{4}$, oe; the RHS must be evaluated, not just stated.
	Assumes true for $n=k$: $\sum_{r=1}^{k+1}\ldots = \log\left(\frac{(2k)!(2k+1)!}{2^{4k}(k!)^4}\right)$ $+\log\left(1-\frac{1}{4(k+1)^2}\right)$	**M1**	States the assumption for $n=k$ and adds the $(k+1)$th term to the assumed sum.
	$= \log\left(\frac{(2k)!(2k+1)!}{2^{4k}(k!)^4}\times\frac{4(k+1)^2-1}{4(k+1)^2}\right)$	**M1**	Combines the two logs into one and writes the extra term as a single fraction. Condone slips in expanding.
	$= \log\left(\frac{(2k)!(2k+1)!(2k+1)(2k+3)}{2^{4k}(k!)^4(2k+2)^2}\right)$	**A1**	Correct single log with the extra term as $\frac{(2k+1)(2k+3)}{(2k+2)^2}$, oe such as $4(k+1)^2$ in the denominator.
	$= \log\left(\frac{(2k+2)!(2k+3)!}{2^{4k}(k!)^4(2k+2)^4}\right)$ $= \log\left(\frac{(2k+2)!(2k+3)!}{2^{4(k+1)}((k+1)!)^4}\right)$	**A1**	cso. Factorial identities for $(2k+2)!$, $(2k+3)!$ and $(k+1)!$ used without error to reach the $n=k+1$ form; working from both ends to a common expression accepted.
	True for $n=1$; true for $n=k \Rightarrow$ true for $n=k+1$; so true for all positive integers n	**E1**	States true for $n=1$, that truth for $n=k$ implies truth for $n=k+1$, and hence true for all positive integers n.
	Total for Question 10	**6**	